REVIEWED FOR
CODE COMPLIANCE
VALIDITY OF PERMIT
The issuance or granting of a permit or approval of plans, specifications and computations shall not be construed to be a permit for, or an approval of, any violation of the provisions of this code or of any other ordinance of the jurisdiction
DATE 1/7/2026
BLDG-2514996

GEOTECHNICAL REPORT

NW Timmerman Road
Forest Grove, OR

For
Grove Vineyards LLC
26 June 2024



EXPIRES:
12/31/2024

Rapid Soil Solutions INC
3915 SW Plum Street
Portland, OR 97219
503-816-3689

1- INTRODUCTION

Rapid Soil Solutions Inc (RSS) has prepared this geotechnical report, as requested, for the proposed development of a new pole building / Agriculture building for the adjacent winery. The parcel assigned the state tax lot identification number of 2N42900-00900 and 2N43200-00400. The site has no address but is opposite Apolloni Vineyards on NW Timmerman Road. The plan is to construction a driveway and a pole building for agriculture proposes.

This report is based on visual observations of the subject site, two shallow hand auger borings, and a review of available literature as referenced at the end of this report. Slopes and disturbance envelopes discussed in this report are approximate, primary based on on-site observations by RSS staff. RSS conducted site investigations unaccompanied.

2- SITE DESCRIPTION

2.1 Location

The subject site is located in rural slopes of unincorporated Washington County between the unincorporated community of Gales Creek and the Hayward area. The Site is situated within the opposite Apolloni Vineyards. Driveway access to the new building will be about 500ft long. The site is accessed off NW Wilson River Highway. The property west of the subject site is undeveloped and is predominantly occupied by wooded slopes, tall grasses and an unnamed creek that the driveway will cross.

The Site can be found within Section 29, Township 2-North, Range 4-West (W.M.) and is within the parcel assigned lot number 900 and 400. The subject property is assigned the county account number of R814079. The latitude and longitude of the site are 45.622648 and -123.219276. The site can be found near the north center of the Gales Creek 7.5-minute quadrangle.

2.2 Slopes

The future pole building will be located on the opposite side of the unnamed stream. A culvert will be placed in the stream and new gravel driveway construction to the pole building. Planned driveway will be at the north side of the property where the slopes are in the 10% range that extend up to the future building. There are no slopes above 25% in the planned development area.

Lidar imagery of the slopes contain some irregulars, suggesting minor debris flows or slumps in the steeply cut slopes at the eroded edge of the meandering stream system. But this area is greater than 400ft from the planned development area and that is only located in the southwest section of the lot and small stream bed.

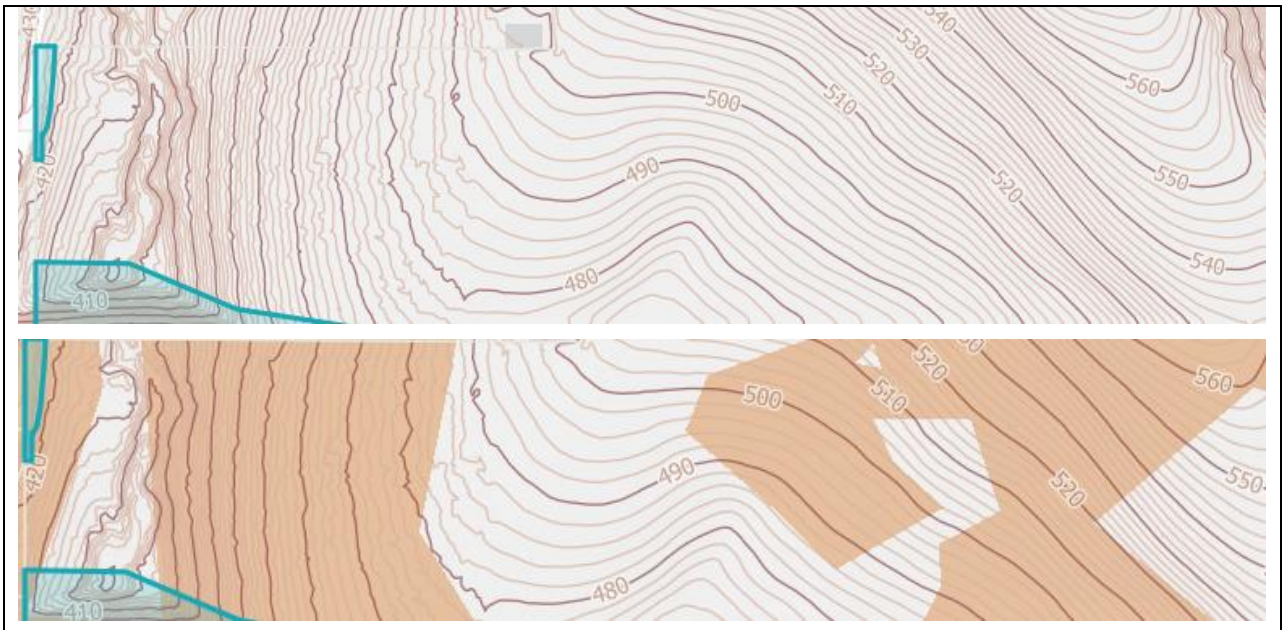


Figure 1: Two-foot contours of the subject site and surrounding slopes. RSS understands that the proposed new dwelling will be constructed around the contour line of 490, which is 500ft from the roadway.

2.3 Built/ Historical Conditions

The site is rural and not developed in any way.

3- GEOLOGY

3.1 Regional Geology

The subject site is situated at the eastern edge of the Oregon Coast Range uplands as the slopes transition into the Tualatin Basin. The Coast Range Mountains form an uplifted belt of land spanning roughly 200 miles along the Pacific Ocean, extending inland roughly 30- to 40-miles. Moderately high peaks averaging 1,500' in elevation, with a maximum of 4,097', can be found in these uplands. The majority of the Oregon Coast Range bedrock is comprised of accreted oceanic sediments and synchronously deposited igneous rocks (where the sediments overlay, underlay and are intruded by the volcanic flows). Uplift, faulting and folding (associated with margin-parallel shortening in the Cascadia subduction zone) lifted the thick stack of sedimentary rock into the heights of the modern mountain range.

Structural deformation has tilted the local bedrock eastwards. The sedimentary beds, which were originally deposited as flat sheets, have been deformed to plunge below the east-adjacent basin. Local mapping suggest that the sedimentary bedrock on the western flank of David's Hill are dipping roughly 39 to 45 degrees in a northeasterly direction. The strike of the local bedrock is approximately parallel to the fault that runs through the middle of the Gales Creek valley.

3.2 Site Geology

The surficial bedrock unit underlaying the Site is classified as part of the Keasey Formation. This unit is comprised of tuffaceous mudstone. It is light grey to buff colored and very thick bedded. Wells et al (2020) notes that the sediments for the unit were derived in large part from the Cascade Volcanic Arc. It was emplaced in a deep marine environment.

The local slopes may contain some thickens of "Gravel of Coast Range provenance". This unit is described by Wells et al as "Diabase and basalt boulder and cobble gravel forming fans and terraces along major rivers and streams draining the Coast Range; as high as 30 m in elevation." West of the site, the valley floor is occupied by relatively young alluvium emplaced by the meandering of Gales Creek.

3.3 Geologic Hazard Document Review

The Oregon HazVu: Statewide Geohazard Viewer and the Washington County Intermap were reviewed June 26, 2024 to investigate mapped geological hazards.

This review indicates that the site is no mapped 100-year floodplain.

The expected earthquake-shaking hazard at the Site is classified as severe (VIII to IX on the instrumental intensity map) with a 10-30% probably of damage from shaking. The site is classified as having a low susceptibility to liquefaction. DOGAMI indicates that the local soils are assigned a D classification on the NEHRP Site Class Map.

Lidar-based landslide mapping, as presented by DOGAMI SLIDO, does not include the presence of slides on or adjacent to the subject property. The nearest slide with located to the east about 1000ft to ¼ miles along the Cedar Canyon Creek River beds side walls

4- GEOTECNIAL INVESTIGATION

A geotechnical investigation was conducted June 16th, 2024. Two shallow hand auger borings were conducted to characterize subsurface soil and collected samples for laboratory analysis.

4.1 Field Exploration Program

The field exploration program for the Site included two shallow hand auger borings. Borings were advanced to the depth of 4 feet. These borings provide the means to characterize subsurface soils and collected soil samples for laboratory analysis.

A professional geotechnical engineer PE GE excavated, observed the borings and logged the subsurface materials and created the soil logs. The logs were created using the Unified Soil Classification and Visual Manual Procedure (ASTM-D 2488). Boring logs are included in the Appendix.

Results of the field explorations are discussed in the Section 5.2.

4.2 Laboratory Testing Program

Two soil samples were collected for laboratory analysis. all samples were analyzed for soil moisture content. One sample was analyzed with an Atterberg limit test. Samples collected for laboratory analysis were transported to the lab in sealed plastic bags.

5- RESULTS OF INVESTIGATION

The following sections present the result of the geotechnical investigation of the Site. Presented conclusions are based on site observations, results of soil characterization, and laboratory analyses

5.1 Site Observations/ Conditions

RSS observed the existing conditions in the subject property, with a focus on the proposed disturbance area. The site is covered with vegetation and has grandual slopes in the proposed area of development.

5.2 Subsurface Conditions

A total of two (2) shallow hand auger borings were conducted at the subject site. The locations are shown in the Appendix. Soil units encountered are summarized in the flowing Table 1. The logs of the hand auger borings are included in the Appendix.

Table 1: Soil Units Encountered

Soil Units		Soil Unit Depths (ft)	
Description	USCS	HA#1	HA#2
Organic rich, silty top soils. Soft to medium stiff, damp, and medium brown in color.	ML (TP)	0-0.5	0-0.5
Medium stiff to stiff, brown, silty CLAY to clayey SILT	ML-CL	0.5-2	0.5-3
Medium stiff to stiff, brown, silty CLAY	CL		3-4

Note: np not present, USCS Unified Soil Classification System Group Symbol (from ASTM D 2487) based on visual observation

5.3 Groundwater

Regional groundwater trends suggest that the groundwater table is relatively deep. No ground water was encounter in the proposed development area.

5.4 Laboratory Tests

The Atterberg limit test tells us the soil is a silty-CLAY soils found across the development area contain low-to-medium plasticity. Moisture contents varied from 22.7% to 26.5%.

6- GEOTECNIAL DESIGN AND RECOMMENDATIONS

6.1 Foundation Design

The building foundations may be installed on either engineered fill or firm native sub-grade that is found at a depth of about 0.5 foot to 1.0feet into the CLAY. This depth may be locally variable and should be confirmed by a geotechnical engineer or their representative at the time of construction. ***Please allow 48hours notice to call for foundation inspections.***

Continuous wall and isolated spread footings should be at least 16 and 24 inches wide, respectively. The bottom of exterior footings should be at least 16 inches below the lowest adjacent exterior grade. The bottom of interior footings should be at least 12 inches below the base of the floor slab.

Footings placed on engineered fill or firm native sub-grade should be designed for an allowable bearing capacity of site 1,500psf. The recommended allowable bearing pressure can be increased by 1/3 for short-term loads such as those resulting from wind or seismic forces.

Based on our analysis the total post-construction settlement is calculated to be less than 1 inch, with differential settlement of less than 0.5 inch over a 50-foot span for maximum column, perimeter footing loads of less than 100 kips and 6.0 kips per linear foot.

Load bearing slabs on grade shall be designed assuming at modulus or subgrade reaction K of 150 *pounds per square inch* (**psi/in**) below grade may be used. There shall be adequate control joints in slabs as to control undesirable shrinkage cracking.

Engineering values summary

Bearing capacity soil:	1500psf
Coefficient of friction soil:	0.28
Active pressure	40pcf
Passive pressure	300pcf

6.2 Retaining Walls and Embedded Walls

Default lateral soil load for the design of basement and retaining walls supporting level backfill shall be 35 psf/ft for laterally unrestrained retaining walls and 60 psf/ft for laterally restrained retaining walls.

For embedded building walls, a superimposed seismic lateral force should be calculated based on a dynamic force of $5H^2$ pounds per lineal foot of wall, where H is the height of the wall in feet and applied at 1/3 H from the base of the wall. The wall footings should be designed in accordance with the guidelines provided in the "Foundation Design" section of this report. These design parameters have been provided assuming that back-of-wall drains will be installed to prevent buildup of hydrostatic pressures behind all walls.

The backfill material placed behind the walls and extending a horizontal distance equal to at least half of the height of the retaining wall should consist of granular retaining wall backfill as specified in the "Structural Fill" section of this report. The wall backfill should be compacted to a minimum of 95 percent of the maximum dry density, as determined by ASTM D698. However,

backfill located within a horizontal distance of 3 feet from the retaining walls should only be compacted to approximately 92 percent of the maximum dry density, as determined by ASTM D698. Backfill placed within 3 feet of the wall should be compacted in lifts less than 6 inches thick using hand-operated tamping equipment (e.g., jumping jack or vibratory plate compactors). If flat work (e.g., sidewalks or pavements) will be placed atop the wall backfill, we recommend that the upper 2 feet of material be compacted to 95 percent of the maximum dry density, as determined by ASTM D698.

A minimum 12-inch-wide zone of drain rock, extending from the base of the wall to within 6 inches of finished grade, should be placed against the back of all retaining walls. Perforated collector pipes should be embedded at the base of the drain rock. The drain rock should meet the requirements provided in the "Structural Fill" section of this report. The perforated collector pipes should discharge at an appropriate location away from the base of the wall. The discharge pipe(s) should not be tied directly into storm water drain systems, unless measures are taken to prevent backflow into the wall's drainage system. Settlements of up to 1 percent of the wall height commonly occur immediately adjacent to the wall as the wall rotates and develops active lateral earth pressures.

6.3 Seismic Design Criteria

The seismic design criteria for this project found herein is based on the ASCE 7-16. A summary of IBC seismic design criterion is below it is generated from the USGS web site for earthquake hazards using a latitude of 45.622648 and a longitude -123.219276, soil site class D,
Null = see section 11.4.8

	Short Period	**1 Second**
Maximum Credible Earthquake Spectral Acceleration	S_S = 0.959 g	S_1 = 0.481 g
Adjusted Spectral Acceleration	S_{MS} = 1.071 g	S_{M1} = null
Design Spectral Response Acceleration Perimeters	S_{DS} = 0.714 g	S_{D1}= null

6.4 Driveway cross section

In order to meet current fire department loading the roadway shall consist of 6" of 1 ½" minus with 2" of ¾" minus on top. RSS will need to proof rolls the excavated roadway with a loaded dump truck to ensure the driveway is hard and non-yielding. Please give 24hours notice when proof rolling. If driveway is constructed during the dry season April to October, geo-textile fabric will not be required.

6.5 Excavations

The initial site preparation will consist of topsoil stripping, and the removal of trees, where applicable. Removal of trees should include removal of the root ball, and any roots greater than ½-inch in diameter.

Excavations can be accomplished with conventional excavating equipment. All excavations for footings and subgrades in the fine-grained silty-clay should be performed by an excavator or backhoe equipped with a smooth-faced bucket (no teeth).

Because of safety considerations and the nature of temporary excavations, the Contractor should be made responsible for maintaining safe temporary cut slopes and supports for utility trenches, etc. We recommend that the Contractor incorporate all pertinent safety codes during construction, including the latest OSHA revised excavation requirements, and based on soil conditions and groundwater evidenced in cuts made during construction.

6.6 Structural Fills

Depending upon finished building pad elevations, structural fills may be required to raise the site grades. Additionally, fill may be required for the backfilling of the proposed new foundation walls. Native or imported material may be used for fill, provided the soil is free of organics, cobbles larger than 6 inches in maximum diameter, or other deleterious matter; is of low plasticity; and, is at the proper water content.

Fills should be placed on level benches in thin lifts and compacted to a dry density of at least 92% of its Maximum Dry Density (MDD) as determined by the Modified Proctor Test (ASTM D-1557).

For any over-excavation completed in the area of footings or slabs, the backfill material shall consist of free-draining, well-graded, crushed aggregate base with a maximum particle size of ¾ inch. The rock shall not contain more than 5% fines (material passing the No. 200 sieve, as tested by ASTM D-1140). The rock shall be compacted to a dry density of at least 92% of its MDD. Compaction tests are required for backfills over two (2) feet and every two (2) feet

6.7 Groundwater Management

The Contractor should be made responsible for temporary drainage of surface water and groundwater as necessary to prevent standing water and/or erosion at the working surface.

The ground surface around the structure should be sloped to create a minimum gradient of 2% away from the building foundations for a distance of at least 5 feet. Surface water should be directed away from all buildings into drainage swales or into a storm drainage system. “Trapped” planting areas should not be created next to any buildings without providing means for drainage. Foundation house drains are required.

6.8 Construction Observation

Prior to pouring any foundation the excavation shall be observed by the Geotechnical Engineer to ensure that the above items have been properly removed. ***Please allow 48-hour notice to call for subgrade inspections.*** Failure to do so can lead to foundation issues with the house. For placement of any backfilling RSS shall be called to provide compaction testing or probing of the fills

6.9 Conclusions

There are no issues with the development of the lot.

7.0 Limitations

This report has been prepared for the exclusive use of the addressee, and their architects and engineers for aiding in the design and construction of the proposed development. It is the addressee's responsibility to provide this report to the appropriate design professionals, building officials and contractors to ensure correct implementation of the recommendations.

The opinions, comments and conclusions presented in this report were based upon information derived from our literature review, field investigation and laboratory testing. Conditions between, or beyond, my exploratory test pits may vary from those encountered. Unanticipated soil conditions and seasonal soil moisture variations are commonly encountered and cannot be fully determined by merely taking soil samples. Such variations may result in changes to our recommendations and may require that additional expenditures be made to attain a properly constructed project. Therefore, some contingency fund is recommended to accommodate such potential extra costs.

If there is more than 2 years time between the submission of this report and the start of work at the site; if conditions have changed due to natural causes or construction operations at, or adjacent to, the site; or, if the basic project scheme is significantly modified from that assumed, it is recommended this report be reviewed to determine the applicability of the conclusions and recommendations. The work has been conducted in general conformance with the standard of care in the field of geotechnical engineering currently in practice in the Pacific Northwest for projects of this nature and magnitude. No warranty, express or implied, exists on the information presented in this report. By utilizing the design recommendations within this report, the addressee acknowledges and accepts the risks and limitations of development at the site, as outlined within the report.

8.0 References

DOGMAI Geologic Map of Oregon (https://gis.dogami.oregon.gov/maps/geologicmap/)
DOGAMI Oregon HazVu: Statewide Geohazards Viewer (https://www.oregongeology.org/hazvu/)
DOMGAI & Oregon Lidar Consortium, Lidar Viewer. (https://gis.dogami.oregon.gov/maps/lidarviewer/)
DOGAMI SLIDO: Statewide Landslide Information Layer for Oregon.
(https://gis.dogami.oregon.gov/maps/slido/)
Google Earth
Google Maps (https://www.google.com/maps/)
Metromap (https://gis.oregonmetro.gov/metromap/)
Oregon Water Rescources Department Well Report Query
(https://apps.wrd.state.or.us/apps/gw/well_log/)
United States Department of Agriculture Natural Resources Conservation Service: Web Soil Survey.
(http://websoilsurvey.sc.egov.usda.gov/App/WebSoilSurvey.aspx)
USGS topographic map viewer (https://ngmdb.usgs.gov/topoview/viewer/)
USGS Earth Explorer (earthexplorer.usgs.gov/)
Washington County, Oregon, maps online.
(https://www.co.washington.or.us/assessmenttaxation/giscartography/maps-online.cfm)

Burns, W.J., and Coe, D.E., 2011, Missoula floods - inundation extent and primary flood features in the Portland metropolitan area, Clark, Cowlitz, and Skamania Counties, Washington, and Clackamas, Columbia, Marion, Multnomah, Washington, and Yamhill Counties, Oregon: Oregon Department of Geology and Mineral Industries, Interpretive Map Series 36

Hart, D.H., and Newcomb, R.C., 1965, Geology and ground water of the Tualatin Valley, Oregon: U.S. Geological Survey, Water-Supply Paper 1697, scale 1:48,000.

Minervini, J.M., O'Connor, J.E., and Wells, R.E., 2003, Maps showing inundation depths, ice-rafted erratics, and sedimentary facies of Late Pleistocene Missoula floods in the Willamette Valley, Oregon: U.S. Geological Survey, Open-File Report OF-2003-408, scale 1:250,000.

Schlicker, H.G., and Deacon, R.J., 1967, Engineering geology of the Tualatin Valley region: Oregon Department of Geology and Mineral Industries, Bulletin 60, scale 1:48,000.

Trimble, D.E., 1963, Geology of Portland, Oregon and adjacent areas: U.S. Geological Survey, Bulletin 1119, scale 1:62,500.

Wells, R.E., Haugerud, R.A., Niem, A.R., Niem, W.A., Ma, Lina, Evarts, R.C., O'Connor, J.E., Madin, I.P., Sherrod, D.R., Beeson, M.H., Tolan, T.L., Wheeler, K.L., Hanson, W.B., and Sawlan, M.G., 2020, Geologic map of the greater Portland metropolitan area and surrounding region, Oregon and Washington: U.S. Geological Survey, Scientific Investigations Map SIM-3443, scale 1:63,360.

Yeats, R.S., Graven, E.P., Warner, K.S., Goldfinger, Chris, and Popowski, Thomas, 1991, Tectonics of the Willamette Valley, Oregon: U.S. Geological Survey, Open-File Report OF-91-441-P, scale 1:100,000.

Appendix

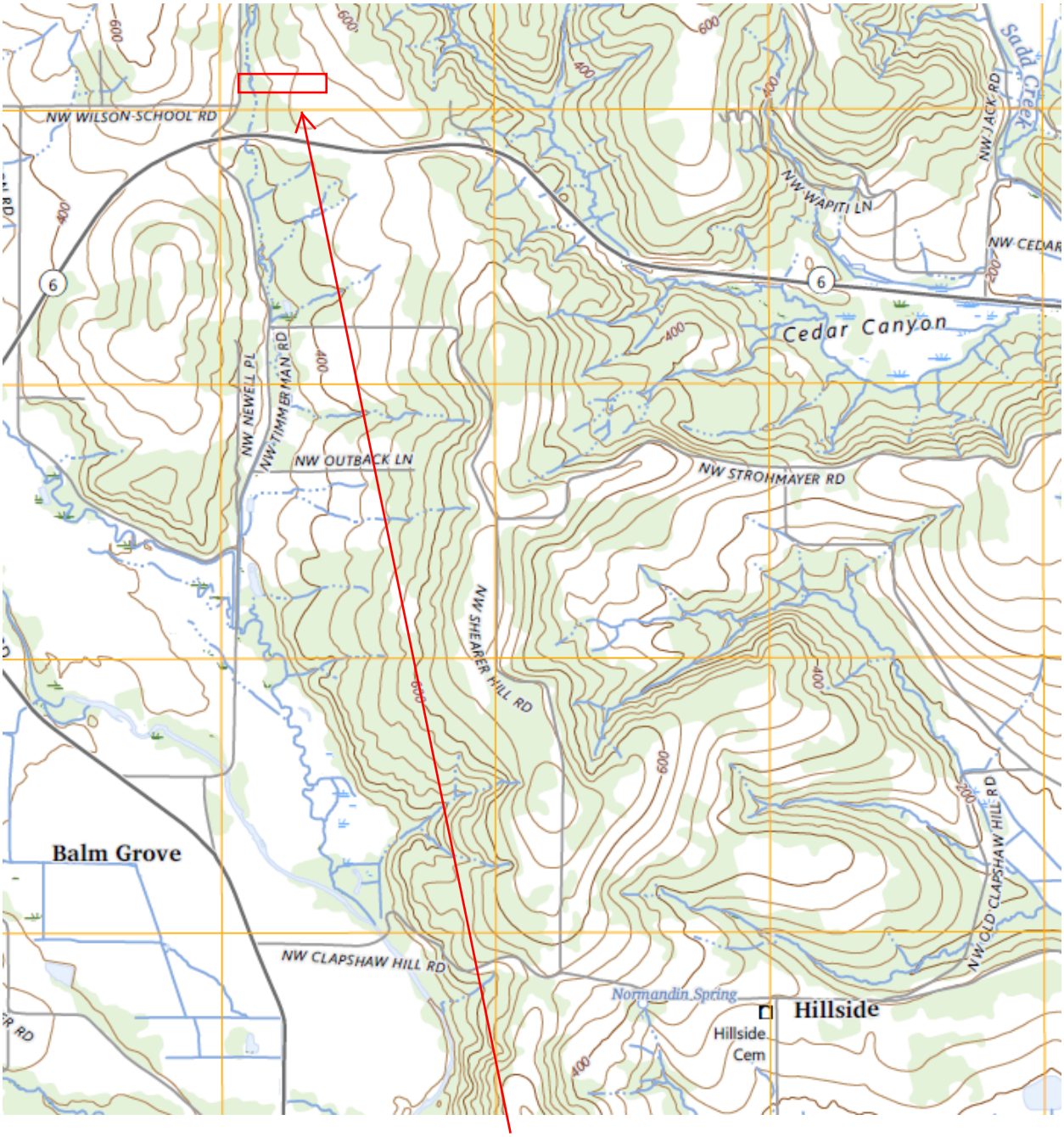


Figure 1 – Site location



Figure 2 – site plan with testing locations

Lab Results

Project Name: NW Timmerman Road Sample Date 6/16/2024

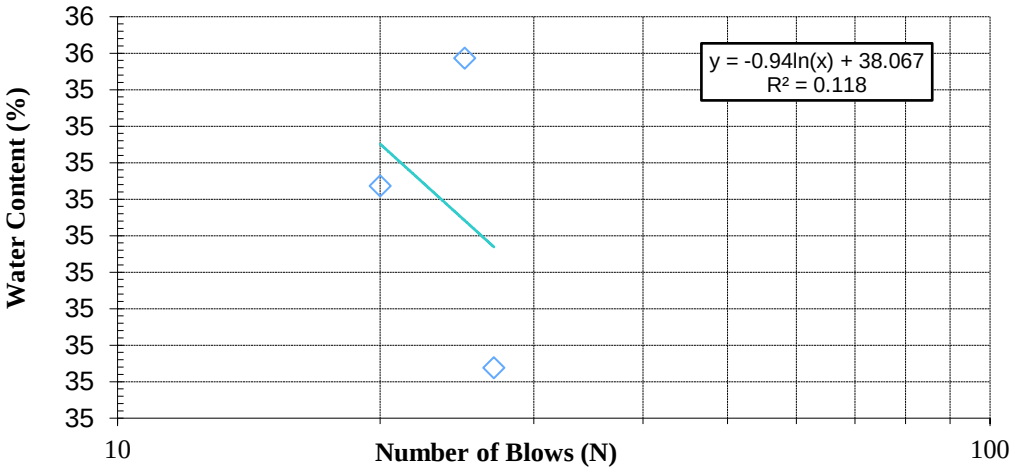
Moisture

	Sample number	HA-1	HA-2			
1	Date and time in oven	6/16 at 10:45AM	6/16 at 10:45AM			
2	Date and time out of oven	6/17- 1:45AM	6/17- 1:45AM			
3	Depth (ft)	2	4			
4	Tare No.	4	5			
5	Tare Mass	232	235			
6	Tare plus sample moist	642	879			
7	Tare plus sample dry	556	760			
8	Mass of water (g)	86	119			
9	Mass of soil (g)	324	525			
10	Water Content (%)	26.5	22.7			

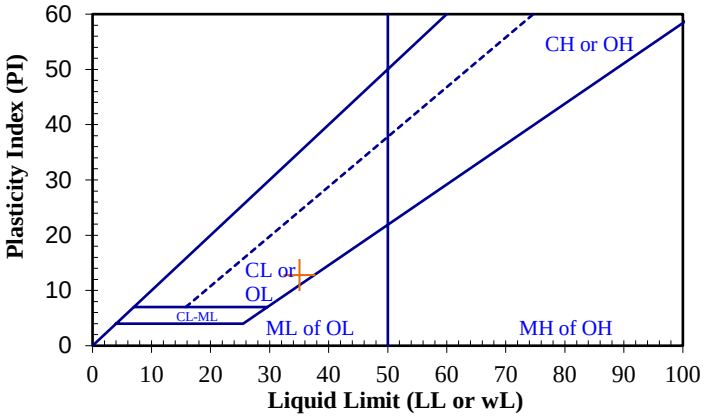
Atterberg Limit Test

Sample Number: HA-1 Depth: 2-ft

		Liquid Limit			Plastic Limit	
1	Tare No.	d1.1	d1.2	d1.3	r1.1	r1.2
2	Tare Mass (g)	39.57	39.93	40.56	39.45	40.25
3	Tare Plus Wet Soil (g)	79.53	103.69	97.66	50.78	50.49
4	Tare Plus Dry Soil (g)	69.14	86.99	82.97	48.80	48.55
5	Mass of Water (g)	10.39	16.70	14.69	1.98	1.94
6	Mass of Soil (g)	29.57	47.06	42.41	9.35	8.30
7	Water Content (%)	35.14	35.49	34.64	21.18	23.37
8	No. Blows	20	25	27		

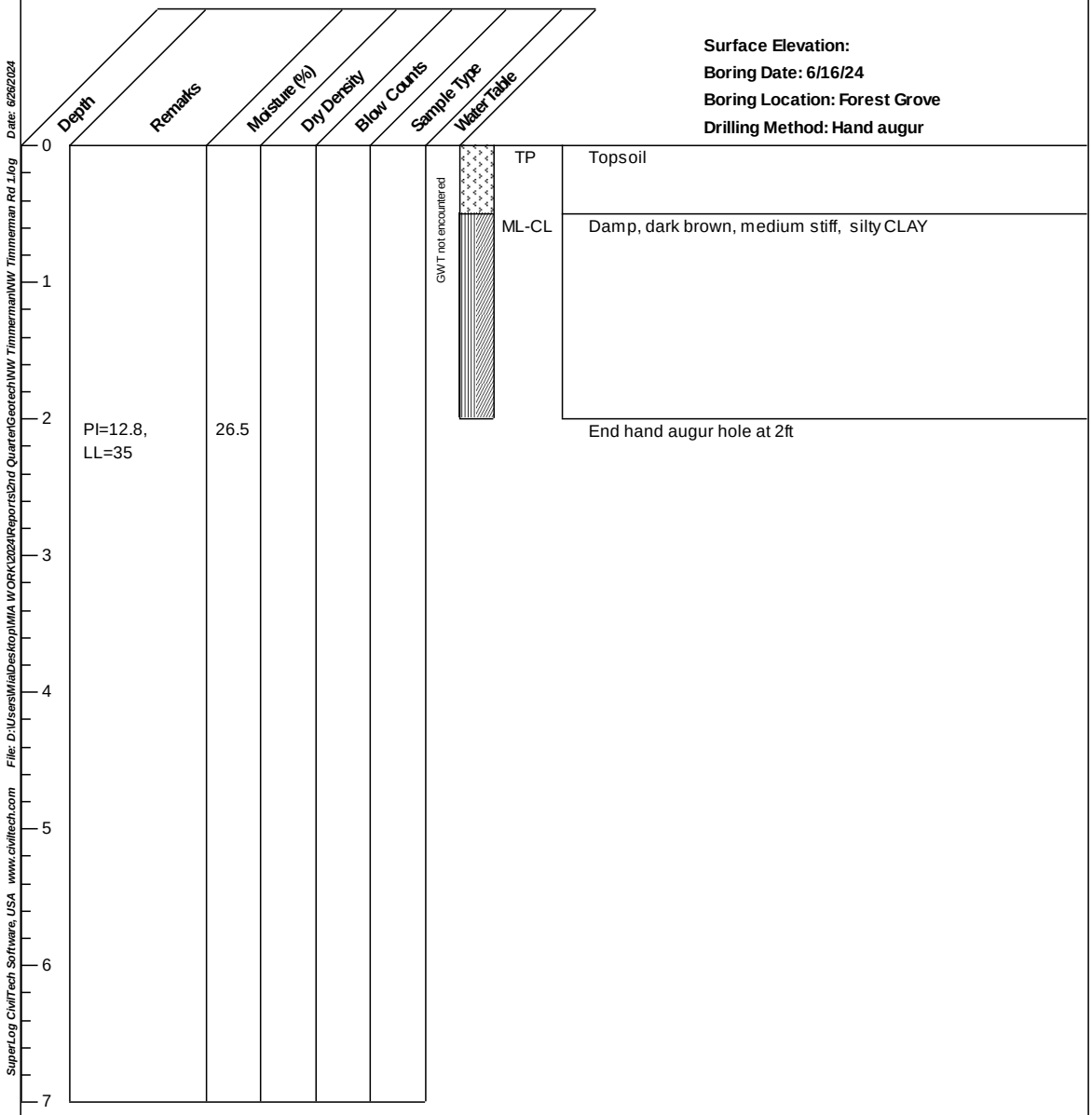


Liquid Limit (%) **35.0**
Plastic Limit (%) **22.3**
Plasticity Index (%) **12.8**
USCS Classification of fines: CL



Rapid Soil Solutions INC

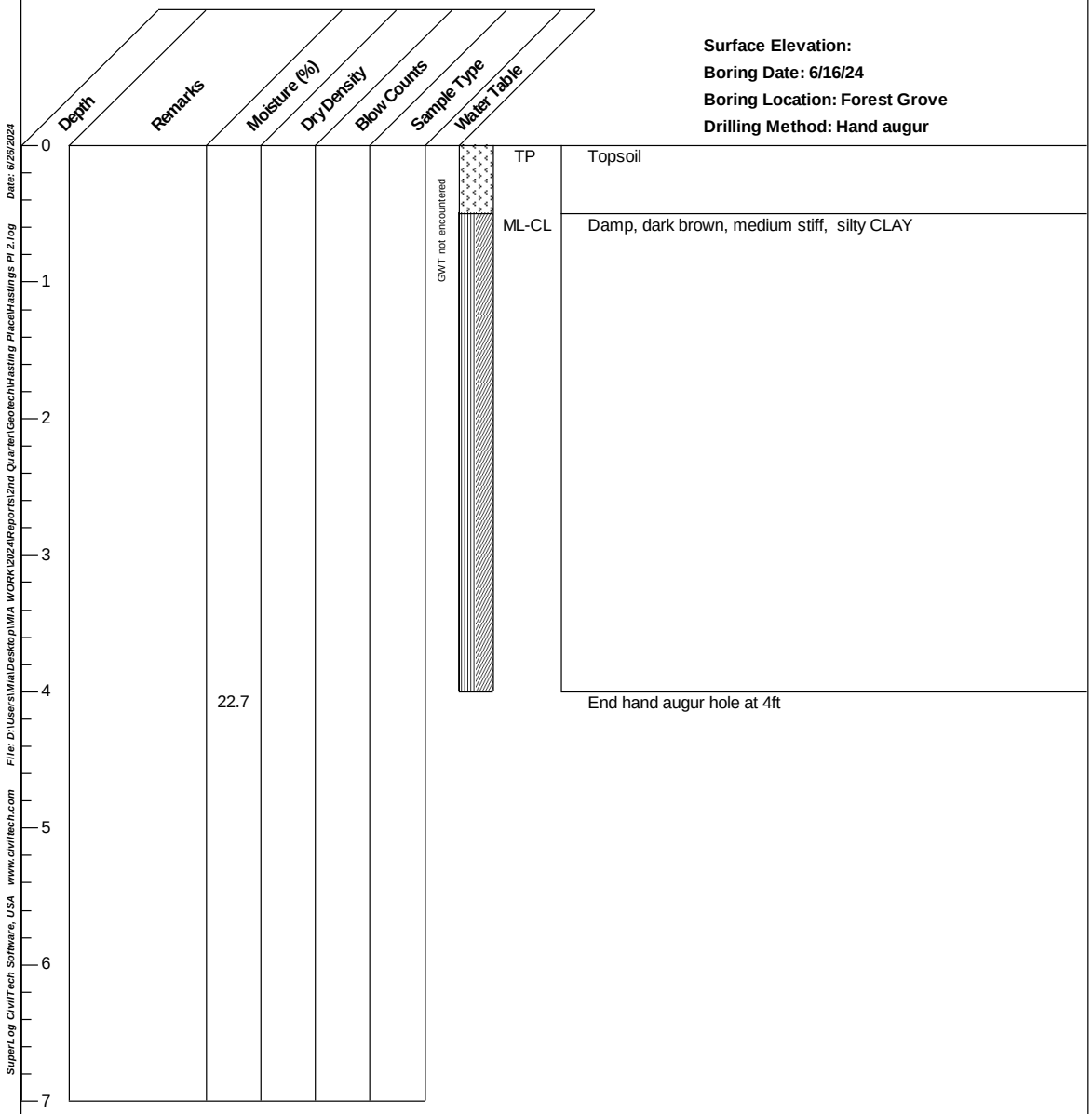
HA#1



LOG OF BORING

Rapid Soil Solutions	**NW Timmerman Rd** **Grove Vinewyard**	**Plate 1**

HA#2



LOG OF BORING

Rapid Soil Solutions	**NW Timmerman Grove Vineyard**	**Plate 1**